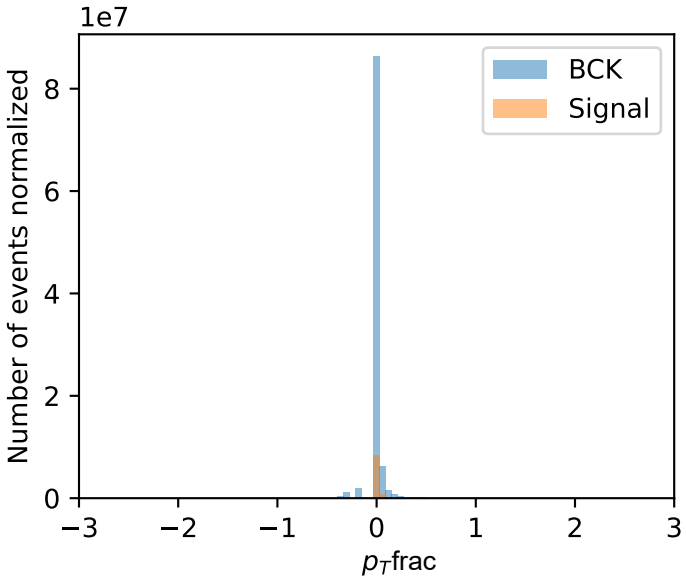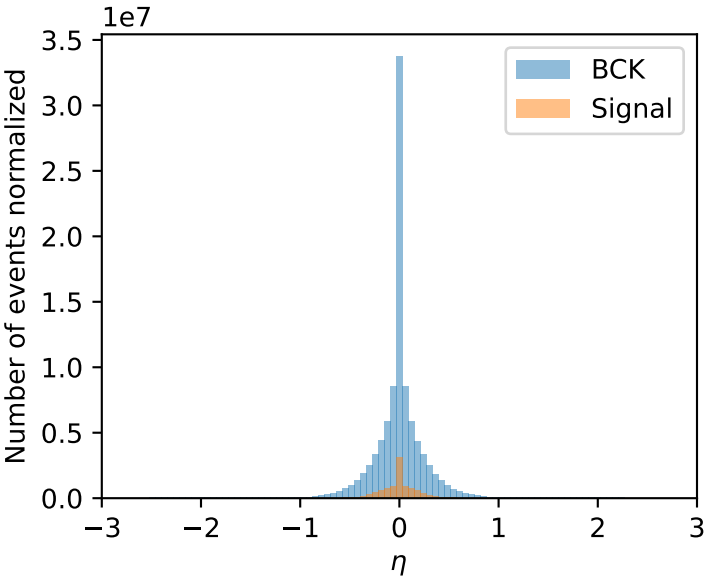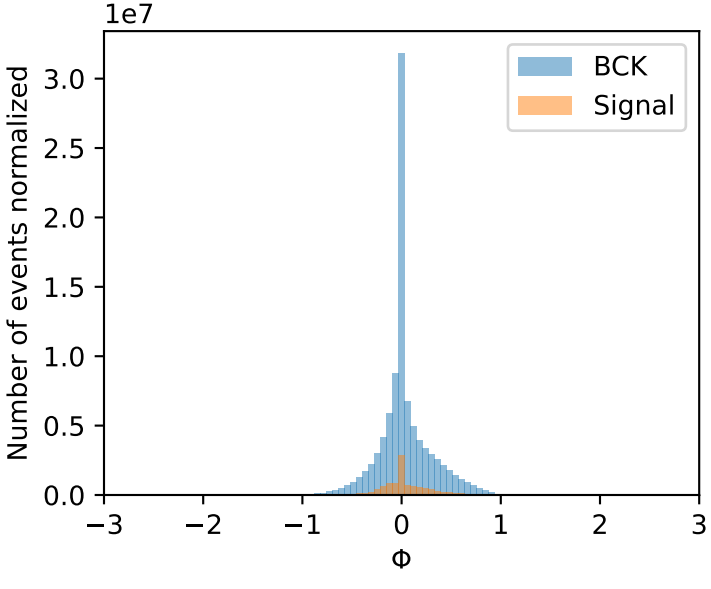Distribution for Normalized reconstructed events